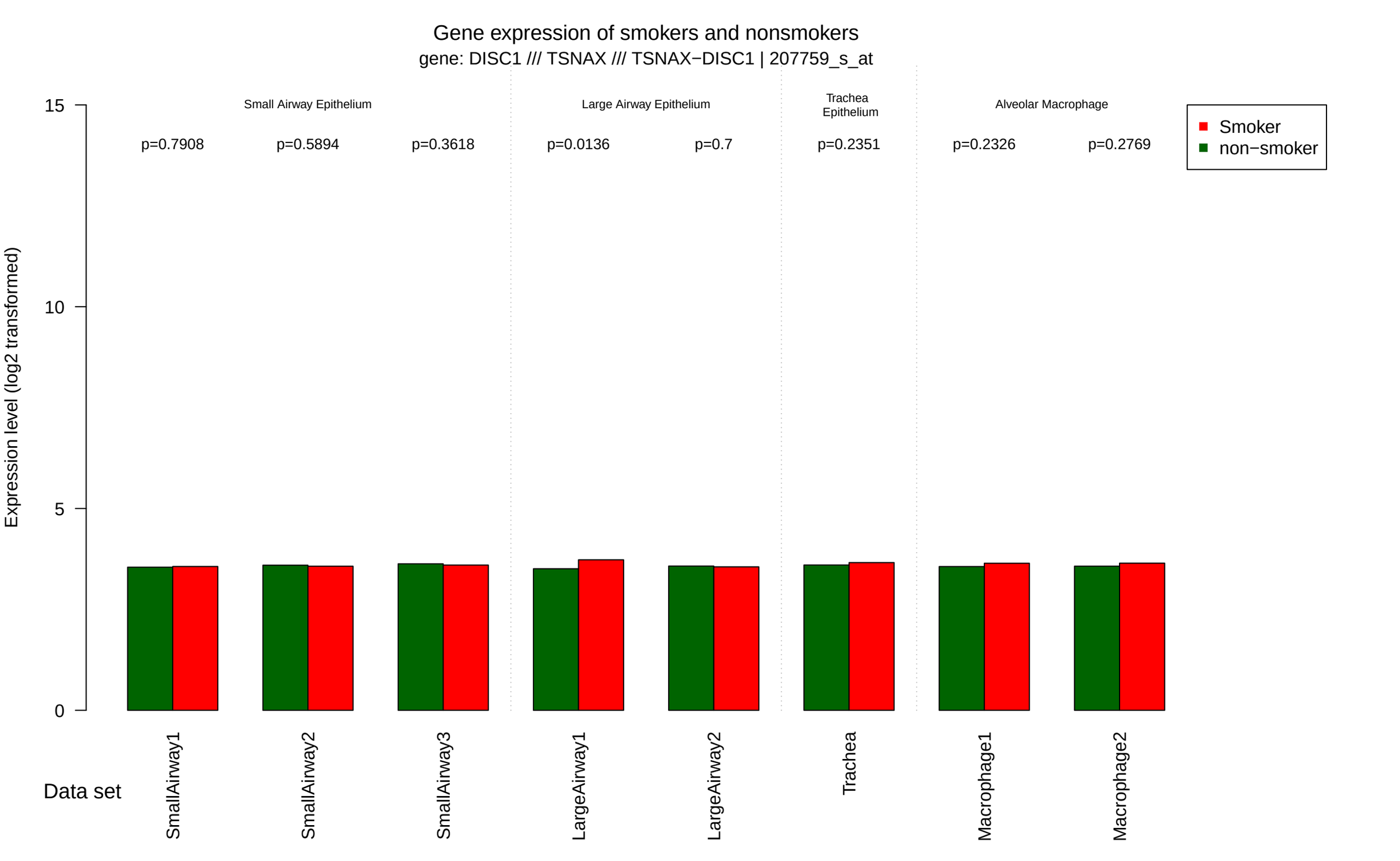
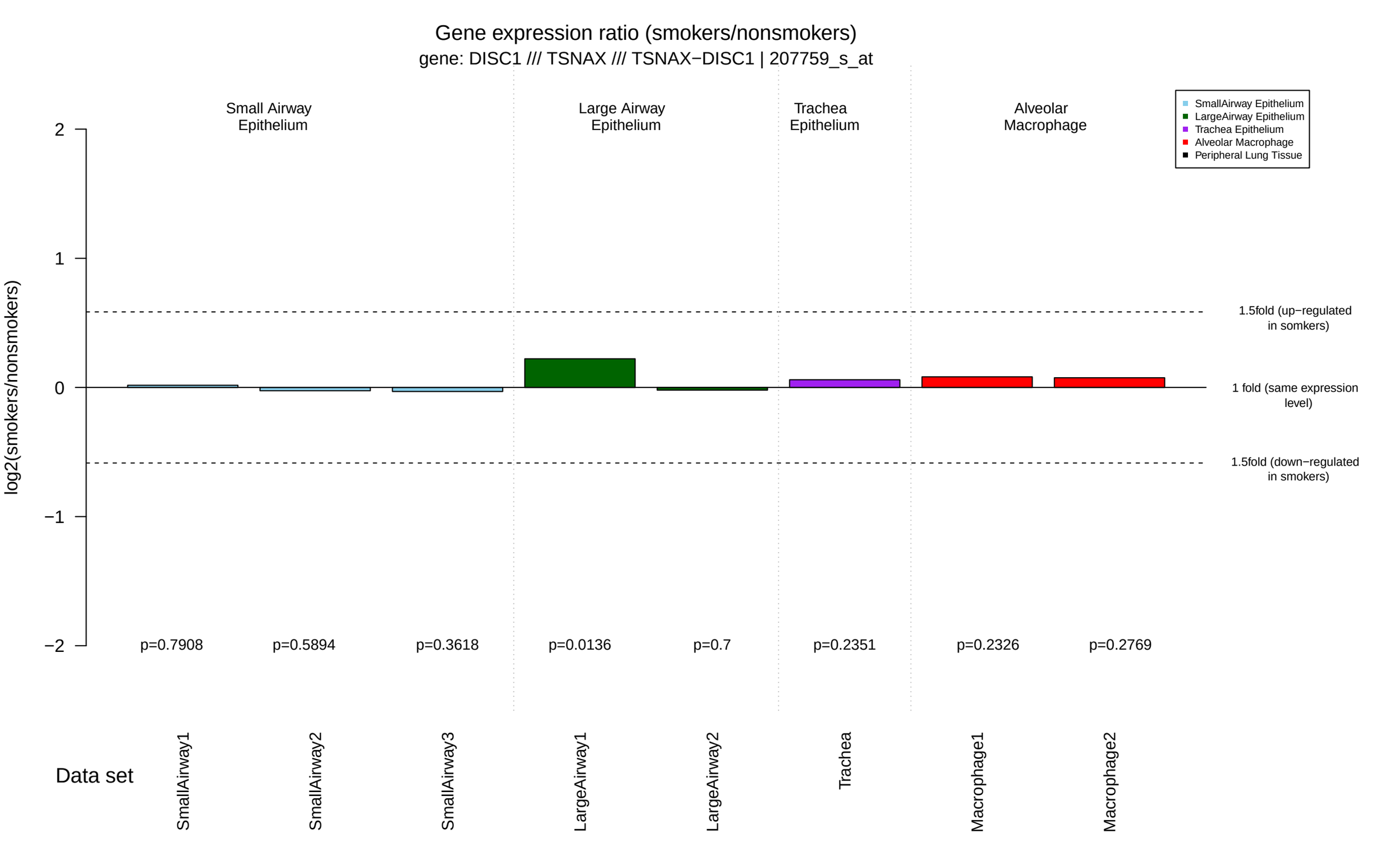
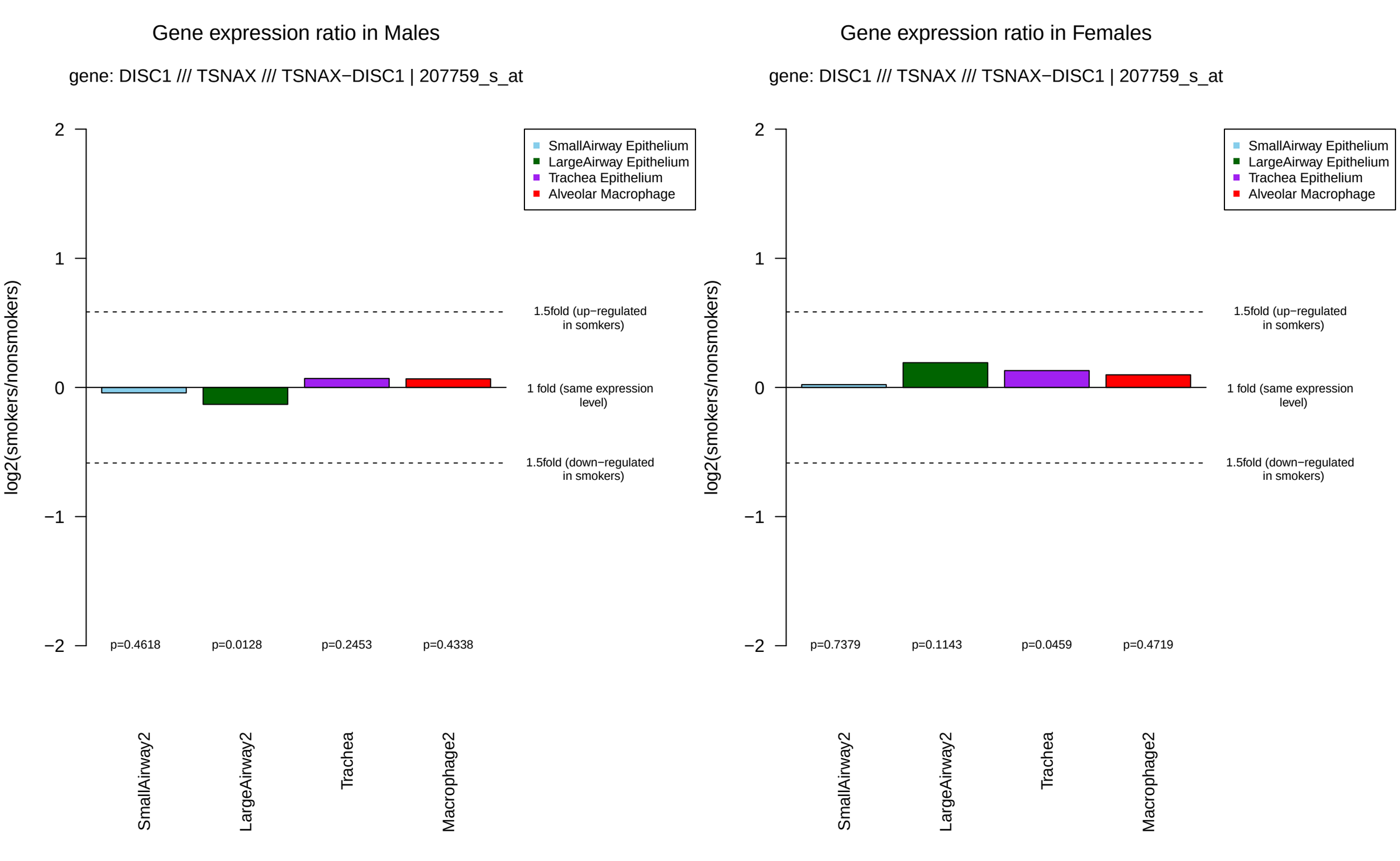
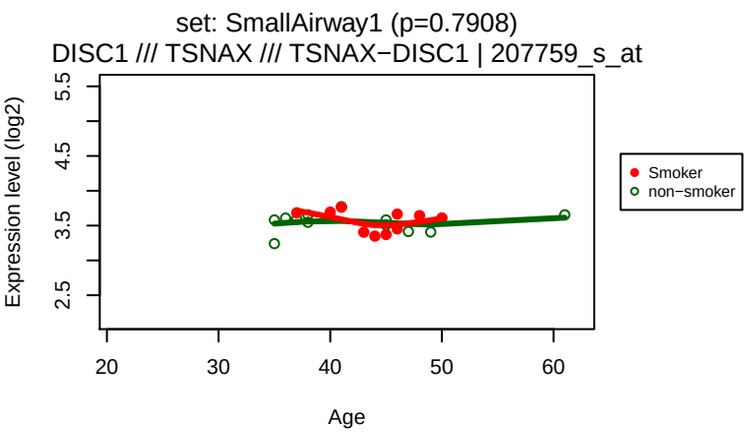
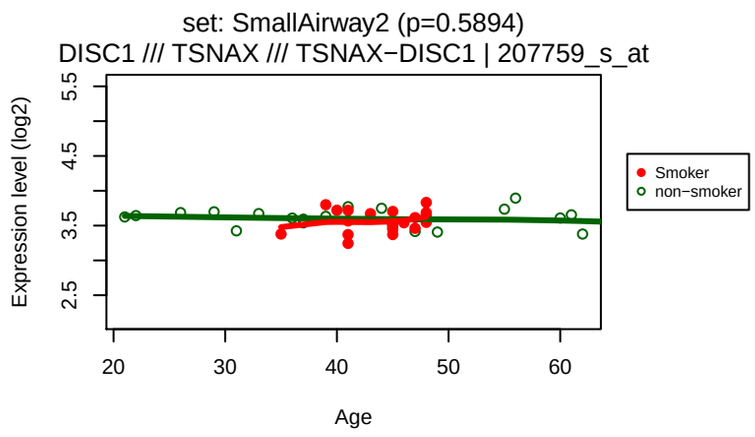
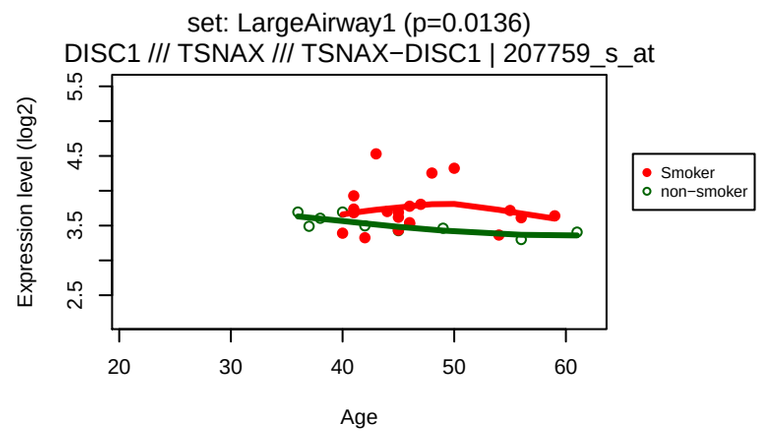

Gene expression of smokers and nonsmokers
gene: DISC1 /// TSNAX /// TSNAX−DISC1 | 207759_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7908
p=0.5894
p=0.3618
p=0.0136
p=0.7
p=0.2351
p=0.2326
p=0.2769
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

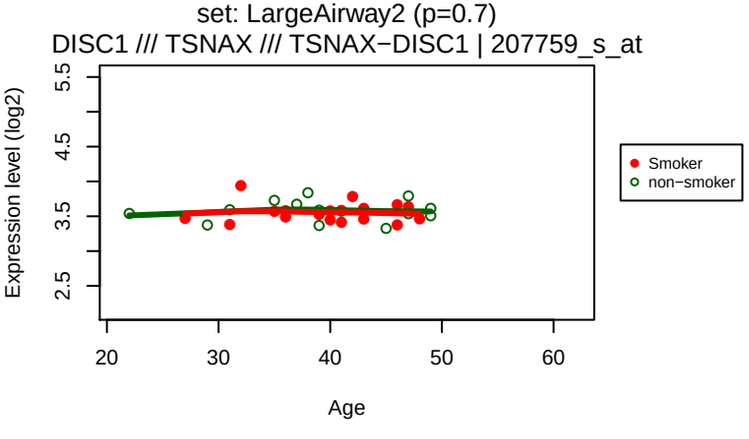

Gene expression ratio (smokers/nonsmokers)
gene: DISC1 /// TSNAX /// TSNAX−DISC1 | 207759_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7908
p=0.5894
p=0.3618
p=0.0136
p=0.7
p=0.2351
p=0.2326
p=0.2769
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

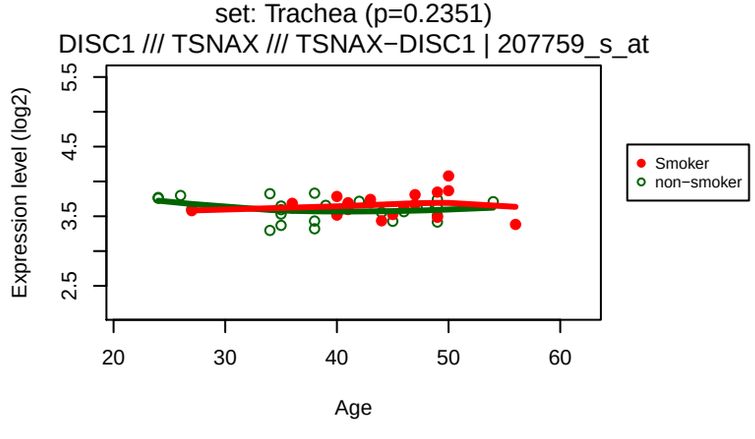

Gene expression ratio in Males
gene: DISC1 /// TSNAX /// TSNAX−DISC1 | 207759_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4618
p=0.0128
p=0.2453
p=0.4338
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DISC1 /// TSNAX /// TSNAX−DISC1 | 207759_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7379
p=0.1143
p=0.0459
p=0.4719
SmallAirway2
LargeAirway2
Trachea
Macrophage2

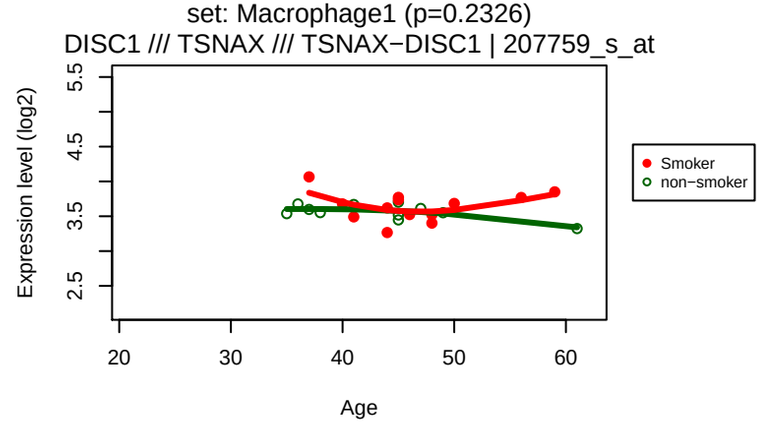

set: SmallAirway1 (p=0.7908)
DISC1 /// TSNAX /// TSNAX−DISC1 | 207759_s_at
Expression level (log2)
Age
Smoker
non−smoker

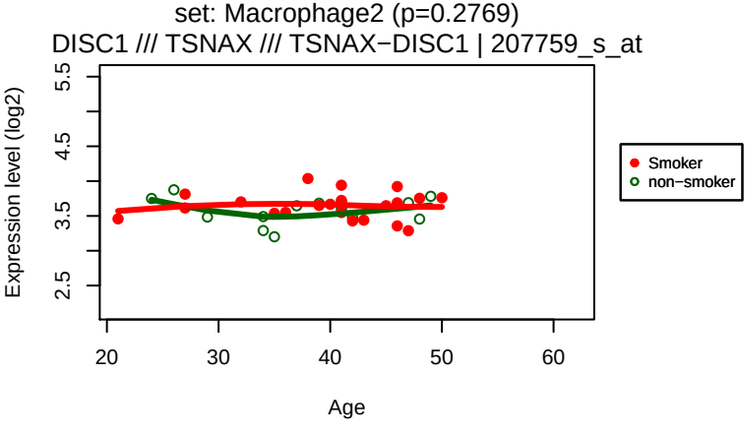

set: SmallAirway2 (p=0.5894)
DISC1 /// TSNAX /// TSNAX−DISC1 | 207759_s_at
Expression level (log2)
Age
Smoker
non−smoker

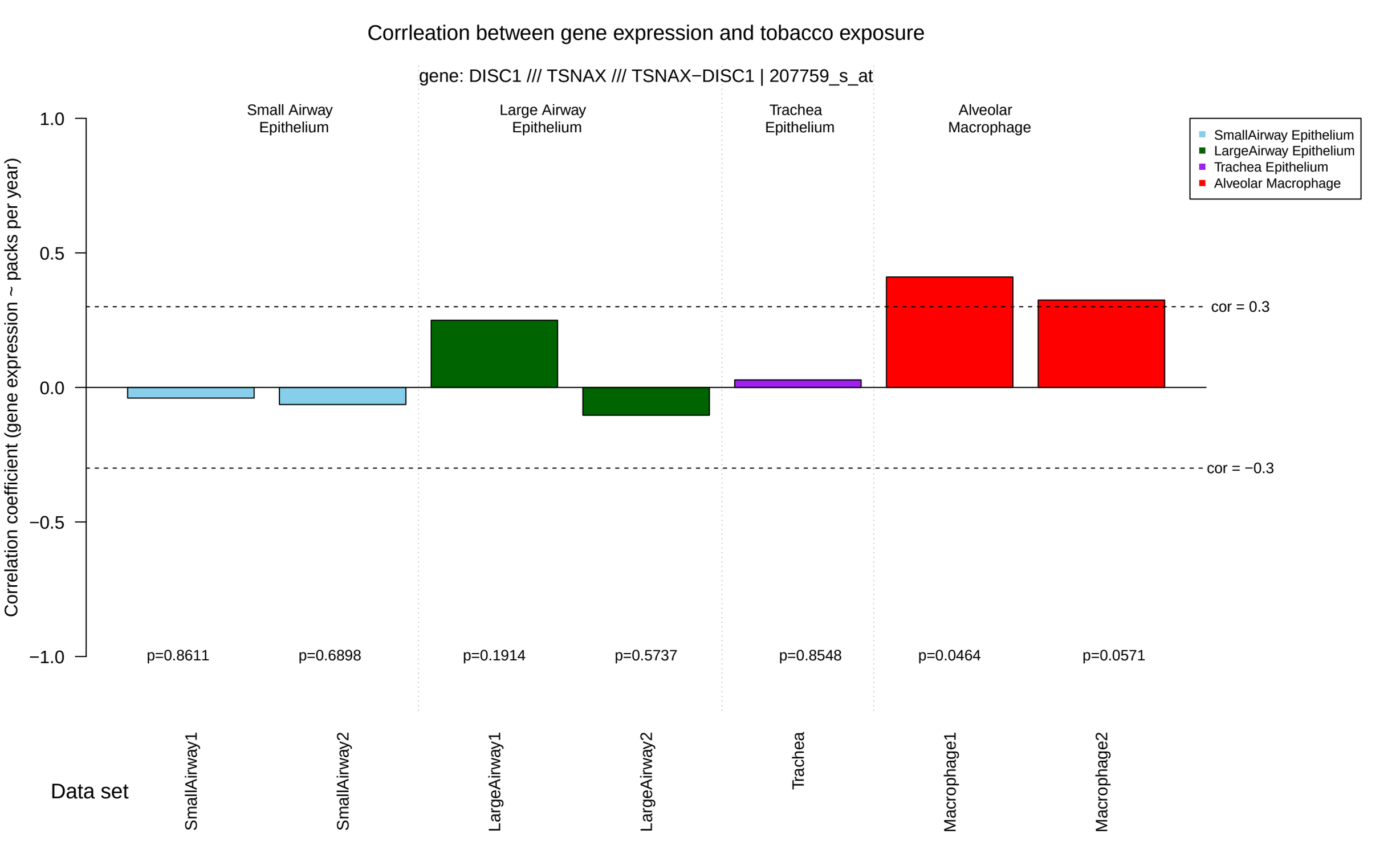

set: LargeAirway1 (p=0.0136)
DISC1 /// TSNAX /// TSNAX−DISC1 | 207759_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7)
DISC1 /// TSNAX /// TSNAX−DISC1 | 207759_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2351)
DISC1 /// TSNAX /// TSNAX−DISC1 | 207759_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2326)
DISC1 /// TSNAX /// TSNAX−DISC1 | 207759_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2769)
DISC1 /// TSNAX /// TSNAX−DISC1 | 207759_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DISC1 /// TSNAX /// TSNAX−DISC1 | 207759_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.8611
p=0.6898
p=0.1914
p=0.5737
p=0.8548
p=0.0464
p=0.0571
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2